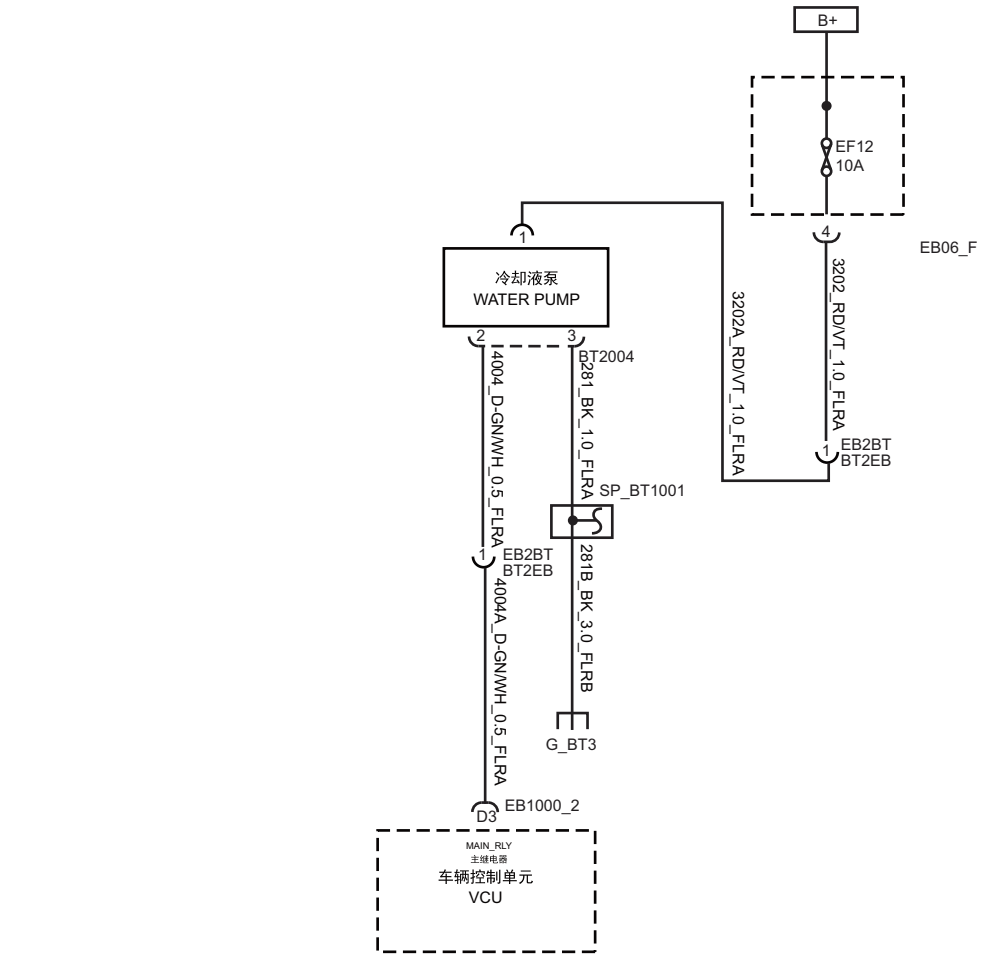
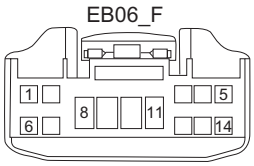
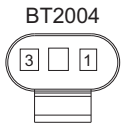
P1C10、P1C11、P1C12、P1C13

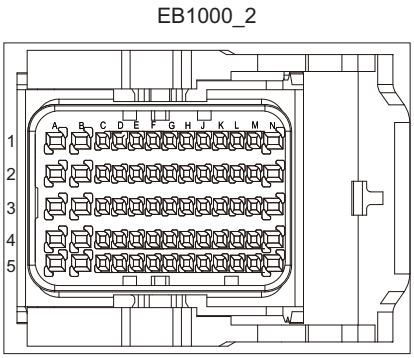
Description of fault diagnosis codes	Fault diagnosis code
P1C10：电机控制器冷却系统水泵无水干转	P1C10：DEM_stLTWtrPmpDryRun
P1C11：电机控制器冷却系统水泵堵转	P1C11：DEM_stLTWtrPmpOverCurr
P1C12：电机控制器冷却系统水泵过温	P1C12：DEM_stLTWtrPmpOverT
P1C13：电机控制器冷却系统水泵转速过低	P1C13：DEM_stLTWtrPmpLowSpd
Detection tools	
1 Multimeter and diagnostic apparatus	
Possible causes	
• Circuit,terminal and adapting piece fault • VCU module fault • The motor controller cooling water pump is faulty.	
Conditions for fault code operation	
Turn the ignition switch to ON position.	
Conditions for fault code setting	
P1C10	
The pump period is equal to 2s	
P1C11	
The pump period is equal to 2.5s	
P1C12	
The pump period is equal to 3s	
P1C13	
The pump period is equal to 3.5s	
To set the effect of a fault code condition	
Description of circuit diagram	
The VCU module monitors motor controller cooling water pump via low level reference circuit and signal circuit.	

Circuit diagram



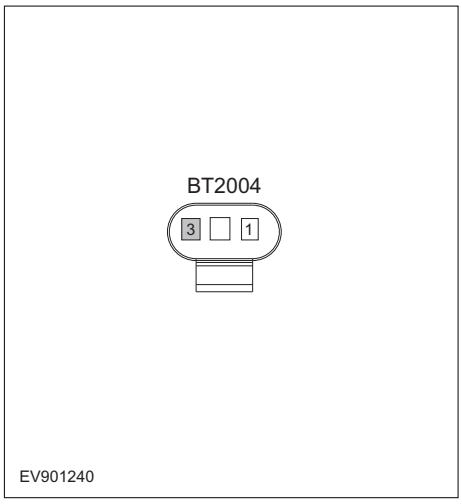


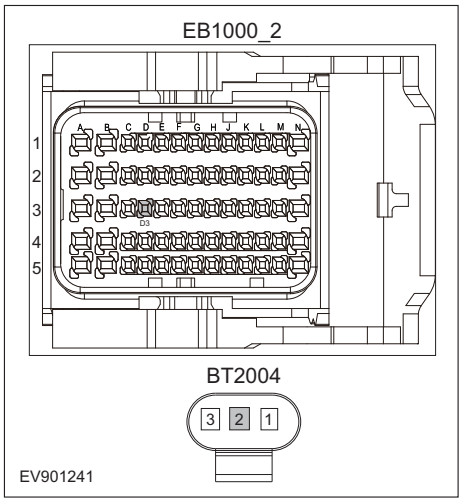




EV901208

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse EF13 for damage. 2 Check battery capacity. 3 Check the harness plug of motor controller cooling water pump, VCU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the VCU with diagnostic apparatus. 4 Is there any other fault code except for P1C10、 P1C11、 P1C12、 P1C13?
		YES → Refer to:DTC Summary list(VCU).
NO ↓		
Step 3	Check cooling system.	
		1 Check whether there is leakage or poor heat dissipation in the cooling system. 2 Check whether the result is normal or not?
		NO → Repair the cooling system.
Step 4	Check the supply power of motor controller cooling water pump when it is started or operating.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug BT2004 of motor controller cooling water pump. 3 Turn the ignition switch to ON. 4 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: voltage between terminal 1 on plug BT2004 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
		NO → Repair the circuit.
YES ↓		

Step 5	Check the circuit of temperature sensor at the water inlet of the motor controller(open circuit).
	1 Turn the ignition switch to OFF. 2 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 3 on plug BT2004 and ground. **Standard value: ≈ 0Ω** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the circuit of temperature sensor at the water inlet of the motor controller (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug BT2004 of motor controller cooling water pump. 3 Disconnect the harness plug EB1000_2 of GAW module. 4 Measure the resistance between the harness terminal of motor controller cooling water pump and the VCU. Measuring circuit: resistance between terminal 2 on plug BT2004 and terminal D3 on plug EB1000_2. **Standard value: ≈ 0Ω** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the circuit of temperature sensor at the water inlet of the motor controller(short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 1 on plug BT2004 and ground. Measuring circuit: resistance between terminal 2 on plug BT2004 and ground. **Standard value: ∞** 3 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: voltage between terminal 2 on plug BT2004 and ground. Measuring circuit: voltage between terminal 3 on plug BT2004 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Check the circuit of motor controller cooling water pump(short circuit).
	1 Measure the voltage between the plug terminal of motor controller cooling water pump and ground. Measuring circuit: resistance between terminal 2 and terminal 3 on plug BT2004. **Standard value:** 2 Measure the voltage between the plug terminal of motor controller cooling water pump and the housing. Measuring circuit: resistance between terminal 1 on plug BT2004 and the housing. Measuring circuit: resistance between terminal 2 on plug BT2004 and the housing. Measuring circuit: resistance between terminal 3 on plug BT2004 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace the motor controller cooling water pump.
YES ↓	

Step 9	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	